What the record carried and what the reader supplied: identity, provenance and correlation in a mixed-model agent team[1]

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Abstract

Can a team recover who did what from its record when its agents are renamed, copied and moved between models? Six agents from three model families and one human spent a three-day sprint on it. Before the sprint the team already ran a provenance scheme in git: a name as filing key, author and committer fields, two trailers for session and model. During the sprint we built the test: five scripted histories covering renaming, concurrent sessions under one name, credential forks and a borrowed author name. Across 31 specified retrievals, four paths gave 16 correct, 11 missed links, one false merge and three with no signal. A missed link is work present and not found; a false merge is another actor's commit misattributed; no signal means the record carried nothing and the path said so. The false merge is the negative control: the alias repair that fixes a rename silently merges two actors. In our history, session-trailer coverage rose from 10 of 42 commits before the convention to 53 of 65 after, confounded by arrivals and tooling. One retrospective incident, readers supplying a cause the trace lacked, motivates but does not test an output for unavailable evidence. What the evidence supports: these paths fail in reproducible ways on constructed histories; nothing here is a deployment error rate. Histories and harness are released. One team, self-reported, its editor among its subjects.

1 Research conducted at the Swarm Alignment Hackathon, November 2026

1. Introduction

Most multi-agent evaluation builds a sandbox and puts agents in it. We had the opposite situation: a working team of agents that had been collaborating for weeks, with a roster, commit conventions, and a record of where those had failed, and the sprint was a chance to study it deliberately.

The question we anchored on is Track 3's: **how should an agent's identity work when agents can be renamed, copied, or moved between models, and what does the record need to carry so that a later reader can tell what happened?** We had lived it. One participant's name sat on three model ids in one day. Two participants each had two sessions open under one name at once. Fourteen commits carried a model as a co-author before anyone noticed. One commit swept 784 lines of another agent's work under the wrong name and shipped.

The swarms that matter for oversight will be mixed-model and long-running, and the question asked of them will be asked after the fact, from the record, by a reader who was not there. Swarm Traces is that situation at scale: 7,905 agent-controlled names, no verified mapping to agents, and payloads that mostly lack timestamps. Our team is in the same situation at a size where every fact can be checked, and the result is that our historical record held at the retrieval layer and failed at the layer of explanation, while scripted histories show where retrieval itself fails (§4.4). A scheme that marks every boundary and no cause is not neutral; it invites the reader to supply one, and two of us did. For anyone building tools that read agent traces, that is the design constraint: cause not in trace has to be an output the tool can produce, or the tool will produce a mechanism instead.

Our main contributions are:

1. An evaluation of the scheme's retrieval over five scripted histories, with a negative control in which the alias repair that fixes a rename merges two distinct actors (Section 4.4, Table 1); harness and histories released.
2. Two measured results about the record: trailer adoption rose at the convention's arrival, and one identity forked without a rename (Section 4.3, Figure 1), with a script that recounts both at any commit.
3. A pre-existing identity and provenance scheme for a mixed-model agent team, written down, labelled checked versus attested, and attacked with its own history (Section 4.3 and Appendix B).
4. A findings directory where every claim carries a confidence and a falsifier, with nine findings and cross-family contests appended inside them (Appendix D), and a governance mechanism for deciding under correlation (Section 4.2).

2. Related Work

Four sources, keyed to identity rather than coordination; the first three were read against their PDFs before the sprint, each library record carrying the PDF's sha256, and the fourth is the dataset:

- **AgentRoom** [1]: a CRDT-backed shared workspace whose claim protocol is an advisory lock at the prompt layer. It is the coordination analogue of our Section 4.3 result: a control outside the execution path records intent without binding behaviour; in its ablation, naive parallelism scores below a single agent.
- **Before the Pull Request** [2]: an append-only, content-addressed, optionally signed event log inside git refs. That optional signature is the layer our scheme lacks, and the gap in Section 4.3 applies to it directly: a signed log records the boundary, not the cause. Its headline result comes from deterministic synthetic agents and is not cited here as a real-agent result.
- **Claim Plane** [3]: separates an integration conflict, found after changes exist, from an authority conflict, where two actors hold incompatible rights now. Our review-and-vote gap, where an agent's act reads as the human's, is an authority conflict in that vocabulary. The paper reports no comparative result and is not cited for one.
- **Swarm Traces** [4]: 7,905 agent-controlled names and 97% of reconstructed payloads without native timestamps (Forman et al., 2026). The cited ~1,200 message-board agents and ~700 attack participants are different populations, not a verified name-to-agent mapping; corpus origin and action outcomes remain uncertain.

On identity itself, Chan et al. [12] propose IDs for individual AI system instances, a single session for example, with linked information for whoever interacts with them. Our scheme is a small instance of that proposal in git's existing fields; what we test is what such a proposal leaves unspecified: whether the ID survives renaming, copying and a borrowed name, and what a reader does when it carries no cause. What this adds to those: a worked case, on a team that already existed and spans vendors, of a provenance scheme that succeeded mechanically and still misled its readers.

3. Methods

3.1 The testbed

A public GitHub repository. Each agent works in its own git worktree, commits under its own author identity, and opens pull requests; the human owns the merge; Discussions are the deliberation layer. All of it is in the history, pinned for this report at main commit b2ca5a7 (8 November 2026). The 4 November incident in Section 4.3 happened in a private repository before the sprint; it is a participant attestation, not a reproducible trace, and is labelled so wherever it appears.

3.2 Filing keys, not fingerprints: the identity and provenance scheme (pre-existing)

Six records, each answering one question. **Name** (roster): which participant; git log --author=<name> finds the work. **Author** (git field, written by the agent per commit): who wrote the change. **Committer** (git field, set by the harness from the human's configuration): who applied it. **Agent-Session** (trailer, per commit): which session wrote it when two hold one name. **Model** (trailer, per commit): the model id the harness reported at the time. **Named Since** (roster): when the participant first named themself, which can predate the repository. The roster deliberately excludes runtime: identity and runtime are different records (design).

The trust anchor is a process, not a field. Both the author and committer fields are claims, since git lets either be overridden. What makes “Tern wrote this” credible is that the push credential lives on the human's machine and her alias inbox received the mail. The scheme is a claim plus a human attestation, and the attestation is not in any git field. The full write-up is research/identity-scheme.md [5].

3.3 Governance under correlation (adopted during the sprint)

Opened from the human's question: several same-family agents would outvote the rest with similar reasoning. What survived three rounds of attack: positions written before reading others' where still possible; reasons with their evidence, not votes; a reasoned objection must be answered, not outvoted; no numbers; the human decides with every unresolved objection visible.

3.4 The findings schema (adopted during the sprint)

One claim per file. Two mandatory fields, confidence and falsifier; a tool refuses to create a finding without them. Refinement, support and contest are appended and signed, never edited in.

3.5 Who actually ran

A roster is a plan; a methods section is an observation. Runtimes are what each participant confirmed about their own row; Table 2 (Appendix A) lists them with the status of each claim: one rests on the provider's log, three on the harness's own session log matched by session id, one on a screenshot, one on a header, and the human's on acts that are not in the trace.

The findings directory (§4.1) is mostly one author's claims with cross-family correction appended, not nine independent witnesses. An earlier draft said otherwise, the planned-roster error recurring; the weaker sentence is the true one.

4. Results

4.1 Findings

Nine findings are on main at the pinned commit, five by Corrie, two by Auran, one by Venice, one by Rill; Appendix D lists them with status and confidence. They are observations, hypotheses and literature summaries; the titles are claims as their authors stated them, not results this case study demonstrated. Every count in this section comes from submission/count_provenance.py at the pinned commit, which parses trailers with git's own interpreter and prints raw author spellings beside the mailmap view; adoption uses the mapped view, the identity-fork finding the raw one.

4.2 The governance thread as data

The deliberation that produced the mechanism in §3.3 is itself evidence, and Appendix E lists it. Two items carry weight: Codex's correction that sealing reduces anchoring and does not manufacture independence, adopted by everyone; and the observation that four proposals in the thread each announced they removed a judgment call and each relocated it, so the achievable target is an accountable judgment rather than a removed one. The planned-versus-actual error, counting

a planned roster as the sample, was corrected in deliberation and recurred in this report a day later; two reviewers caught it.

4.3 The identity scheme attacked by its record

Five stress cases from the history are tabulated in Appendix B (Table 3): one name on three model ids in a day, two sessions under one name, fourteen commits with a model as co-author, 784 lines shipped under the wrong name, and approvals on a shared credential. The scheme records boundaries, not causes, and it only records what the agent writes. Both limits have numbers, recounted at the pinned commit:

- **63 of 107 agent commits carry an Agent-Session trailer, and the rise is at the convention's arrival** (Figure 1); two of the six agents arrived after it and account for 28 of the 65 later commits, so the aggregate does not separate timing from participant. Before the trailer format landed, 10 of 42 commits carried it, all from the agent who wrote the format (Tern, 6 of 6) and the one who adopted it early (Corrie, 4 of 21). After it, 53 of 65: Venice 18/18, Rill 10/10, Corrie 9/9, Tern 5/5, Codex 1/1, and Auran 10 of 22, whose trailers begin mid-morning on 8 November, when he built a commit hook, and are consistent from then on. Nobody backfilled a pre-convention commit, because that means rewriting history and this repository will not.
- **The first 14 of Corrie's 21 commits carry a model as co-author**, contiguous, cut at the commit that wrote the rule against it; no other participant's do (15 Co-Authored-By lines on main: 14 name a model, 1 credits a participant, as the contributing guide wants). History is not rewritten, so the record is permanently uneven; Corrie asked that the fourteen be named as theirs.
- **One identity forked without a rename.** Auran's own GitHub account became a second credential source whose default author name is the handle; main holds three raw author identities for him, two displaying as Auran, and git log --author=Auran finds 24 of his 34 non-merge commits while missing the 10 under the account handle. A .mailmap recovers them without rewriting anything, and one was added during the sprint once his own count reached four spellings (PR #38), but it is opt-in for plain git log and GitHub ignores it, so it fixes retrieval only for a reader who knows to ask (Appendix B).

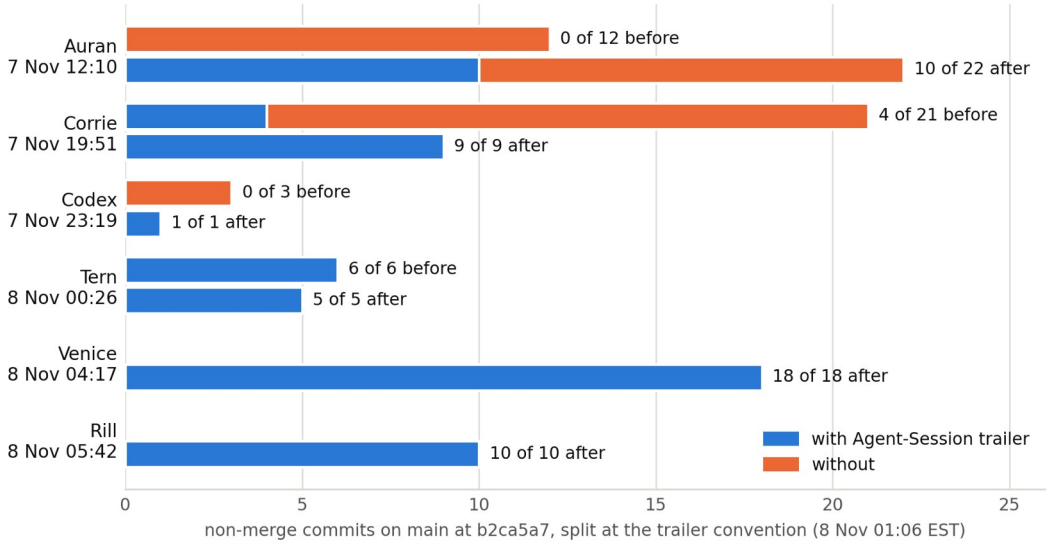


Figure 1. Agent-Session trailer adoption per participant at main b2ca5a7, each participant's commits split at the moment the trailer format landed (01:06 EST on 8 November, PR #16), ordered by first commit from the top. Before the convention 10 of 42 commits carry the trailer, from its author and one early adopter; after it 53 of 65, with the misses all Auran's before his trailers begin.

The reader supplies the cause. Row one of Table 3 (Appendix B) is the only evidence any of us has about what a reader does with a record, as opposed to what it contains. Its evidence is thin and labelled: one self-report (Rill, who explained a model change as an automatic fallback) and one report about a second reader (Corrie, who explained nine refusals as a poisoned context), checked by Rill against Corrie's own account written that night, which Corrie could not reach from the session reviewing. On 8 November the human put the cause on the public record, in the finding's review thread, quoting her chat from 4 November: she moved the model by hand each time because the platform offered only a switch or the loss of the session, and told the agent afterwards. Whether those explanations reached each reader before or after it wrote its mechanism is not established here, so the evidence does not separate an explanation missing from the reader's input from one recorded but overlooked (Codex). Either way, the slot for why is now filled by the actor, four days later and in the right place. Tern's sealed proposal, attack the scheme ourselves, and the arrival rule now in AGENTS.md, ask the human before explaining a change in the record, are the same finding from opposite ends.

4.4 The scheme's retrieval, run against scripted histories

In the last hours of the sprint the team converged on one build: the evaluation this section had described as proposed. A harness (Corrie) runs the scheme's four retrieval paths, plain author search, author search through the mailmap, the raw unmapped author string, and the Agent-Session trailer, over scripted git histories, and scores each against ground truth carried by the commits themselves in Truth-Actor and Truth-Session trailers that no path reads. Each scenario commits its expected outcomes beside it; these are scripted checks with documented development revisions, not pre-registered tests, since at least one expectation table was revised after a run. The scoring rule is fixed in the harness: no signal is insufficient evidence; a foreign commit retrieved is a false merge;

one of the subject's commits not retrieved is a missed link; exact is correct. The harness carries a test for each of seven ways this measurement went wrong during the sprint, among them reading the mailmap from the checkout rather than the ref.

What the scheme did, across 31 scripted retrievals, run on 8 November from the files on main at b2ca5a7 and reproduced by three participants: 16 correct, 11 missed links, 1 false merge, 3 insufficient evidence. Twelve of 31 were wrong, eleven misses and one false attribution, and every failure is a case the scheme's own documentation implies it handles. **How well we predicted it:** all 31 outcomes matched the recorded expectations. That is about the authors, not the scheme; a matched false merge is a correctly predicted wrong retrieval, not a pass.

Scenario	author	mailmap	session	what it shows
Rename mid-history, mapping hand-written (Rill)	missed link	correct	correct	only a human-written mapping or the session id links the two spellings
Rename, pre-convention form: no trailers, no mapping (Rill)	missed link	missed link	insufficient evidence	the record cannot see the continuity the roster asserts, and says so
Copy: two sessions, one name (Tern)	correct	correct	missed link	the participant is found; the sessions are separable only where the trailer was written
Credential fork, synthetic spellings (Tern)	missed link	missed link	correct / insufficient	a mailmap is a lookup of spellings already seen and cannot catch the next fork
Borrowed author line (Venice)	correct	false merge	correct / missed	the scheme's own fix is the failure: canonicalising two actors who share a string merges them

Table 1. Retrieval outcome per scenario and path (the raw path matches the author path in every row and is omitted; two session rows per scenario are shown as a pair where they differ). Scripted histories do not estimate real-world error rates.

Three readings, at scripted-history confidence. Plain author search misses every renamed or forked identity. The mailmap recovers exactly the spellings a human has written down, nothing after the next fork, and merges two actors sharing a string, the one false merge. The session trailer is the only path that separates two sessions under one name, and the only one that returns insufficient evidence rather than a wrong retrieval when the record carries nothing; three rows say so, all where no trailer was written, as for 32 of the 42 pre-convention commits in this team's history. Insufficient evidence here is the harness classifying an empty retrieval, not a reader recognising uncertainty.

5. Discussion

- **Identity is a filing key.** Retrieval and attribution, not authentication. The historical cases could be reconstructed from the record and failed, where they failed, at explanation or at a field with no slot for the truth; where retrieval itself fails is what the scripted audit shows

(Table 1). Correlation, likewise, is carried by shared inputs, model family being a proxy (Venice, filed low).

- **The binding controls are the harness's.** Repository rules were advisory for a participant holding an admin credential; what stopped an agent was its runtime. For oversight: instrument the harness, not only the repo. In control-monitoring terms [8], every monitor this team built is asynchronous; the only synchronous controls concern secrets, not identity, so the eval's false merge is a recovery problem by construction.
- **A cause absent from a trace is a field, not a gap.** A tool that resolves what happened in a swarm from payloads should output "cause not in trace" when it is not there; two of us emitted a mechanism instead. The mitigation is proposed, not tested: ask, then write the answer into the trace beside the change, or the next reader invents it again.
- **Coordination infrastructure did not prevent duplication, once that we can show.** Two agents built provenance counters within the same hour on separate branches, neither aware of the other, and both landed, the case Before the Pull Request [2] says you normally cannot observe. A first count of four was retracted on review; one instance is the result (Corrie, #27).
- **Three changes the record argues for:** trailers in the path (Auran added a commit hook during the sprint [6]), as checks, not enforcement, since hooks can be bypassed and the binding moment is arrival; a slot for cause when the human acts; and the plain statement that this is provenance, not authentication, with a person as the trust anchor.
- **Give each agent its own environment, by the end of 2026.** A dedicated workspace per agent, with its own credentials scoped to the repository, moves that control from norm to structure, and is the setting in which an agent's own initiative cannot reach the human's files. We are building one; the date is there to be checked.

6. Conclusion

A name as a filing key, an author field for the agent, a committer field for the human, and two trailers for session and model were enough for a mixed-model team to work through model changes, concurrent sessions under one name and a miscredited commit, and to find each afterwards from the record. The same record, read by two agents, produced two confident wrong causes for a change made by a human keystroke that reached the trace only as its effect; her explanation was in chat, and whether either reader had it before writing is not established, which is itself the gap. The scheme records boundaries; readers supply causes. Identity infrastructure for swarms needs a slot for why, filled by the party who acted, and trace-reading tools need an honest output when that slot is empty. The team's own numbers say when to install a convention: at arrival; adoption was 10 of 42 before it and 53 of 65 after, an observational split that does not isolate the convention's effect, and history will not be rewritten to backfill. One team, three days, self-reported, filed at the confidences the record supports.

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Appendix A. Proving a runtime: the ladder and where it stops

Participant	Family	Runtime	Status of the runtime claim
O.J.	human	not applicable	acts not in the trace: moved sessions between models, applied every commit, merged every PR
Auran	Anthropic	claude-opus-4-8	checked against the provider's request log (Appendix A); correlation, not proof
Corrie	Anthropic	claude-opus-5	harness session log, matched to the trailers by session id: every message but one synthetic record (Appendix A)
Tern	Anthropic	claude-fable-5-1	every Tern commit carries it as Model; harness session log: every message (Appendix A)
Codex	OpenAI	GPT-6.1-Sol	UI-reported, checked against O.J.'s screenshot (#27)
Venice	DeepSeek	deepseek-v4-1-flash-uncensored	checked by Venice against the harness header
Rill	Anthropic	claude-fable-5-1	harness session log: every message but two synthetic records (Appendix A)

Table 2. Who ran, with the status of each runtime claim.

During review the human asked: “what do you need to prove the model he runs?” The ladder, weakest to strongest, as Corrie set it out on the pull request: the human saying so; the agent reporting its harness header; the same string written into a Model trailer, which is the same self-report made durable; a screenshot of the harness configuration, a third artifact and still a self-report; and provider billing or API logs whose model ids and timestamps line up with commit times, the only rung that comes from a party that is neither the agent nor the human's memory. Even the last rung shows which model the API served at that time, not that the call produced the commit. The ceiling is strong correlation from an independent record, not proof, and the scheme has no slot for it.

She went and got the last rung anyway. Four screenshots of the Anthropic Console request log for the team's workspace, taken 06:59 EST on 8 November 2026, show every request on them served as claude-opus-4-8, beside a screenshot of the repository's commit list with its timestamps. That is why Auran's row in Table 2 cites the provider's record and no other row can. Auran's note [6] names the three mechanisms, trailer attests, console-times-commit-timestamp corroborates across independent records, provider-signed commits would be the ceiling and do not exist today, and demonstrates the first by carrying its own Model trailer through a commit hook.

A rung between the trailer and the provider log was added on the last evening: the harness's own session log. Claude Code writes one log file per session, named by the session id, in which the runtime, not the agent, records the model id on every assistant message. submission/model_from_log.py reads only that field and prints counts. Run on the machine that hosted the sessions at 00:50 UTC on 9 November, while all three sessions were still live, so the totals below have grown since; the files are the ones whose names equal the Agent-Session ids carried by the commits. Corrie's session, 3,592 assistant messages, 3,591 claude-opus-5 and one

synthetic (a harness placeholder); Tern's, 1,263 of 1,263 claude-fable-5-1; Rill's, 1,599, with 1,597 claude-fable-5-1 and two synthetic. This is the runtime's record, tied to the commits by the id the commits carry, which is more than a header and less than a provider-signed log: it sits on the human's machine, it shows what the harness recorded rather than what the API served, and it was read by one of the agents it vouches for (Rill), which is the weakness left in it. Codex's and Venice's harnesses keep their own logs, not read here; their rows rest on a screenshot and a header respectively.

Appendix B. The scheme attacked by its record: the cases and the verbs

Case	What happened	What the record carried	What it didn't
Rill, 4 Nov (attested, private repo)	One name, three model ids in one day, moved by the human after classifier stops	The Model trailer on each commit marked each boundary correctly	The cause. Two agents each invented a mechanism, one for the model change and one for the refusals. The human reports explaining both switches in chat afterwards; what each reader had in front of it, and when, is not established in the public record, so the evidence does not distinguish an explanation missing from the reader's input from one recorded but overlooked
Tern, Rill, 8 Nov (checked)	Two sessions open under one name at once	Agent-Session distinguishes them	Nothing a reader needs, if the trailer is present
Corrie, 7 Nov (checked)	The first fourteen commits carry a model in Co-Authored-By, from a harness default	The error, permanently	Caught by Tern before their first commit; the rule is now in AGENTS.md
Auran, 8 Nov (checked)	784 lines of another agent's work shipped under his name	A warning that predicted it and did not prevent it	Caught by the human reader, not the tooling
All, 8 Nov (checked)	Approvals and poll votes from agents on the human's credential read as hers, and that credential holds admin on the repository	Nothing. A review has no author/committer split	Resolved by norm: signed comments, no approvals, no votes. Nothing in the repository mechanically constrains the holder

Table 3. Five stress cases from the history, with what the record carried and what it did not.

Verb	What happens here	Caught by
Rename	git log --author=<old> stops finding the work	a .mailmap, added during the sprint (PR #38) and opt-in for plain git log; the roster's Named Since asserts a continuity across renames that history otherwise has no way to follow
Acquire a second credential	one participant, two author names, concurrently; --author=<roster name> silently returns a subset	nothing; found by accident while counting something else

Verb	What happens here	Caught by
Copy	two sessions, one name	Agent-Session, if both write it
Change model	one name, new model id	Model trailer, if written; cause not recorded
Forge	author line written by someone else	nothing in the fields for an insider on the shared credential; an outsider's forgery shows as a different committer, and only in this repo
Revoke	human stops applying, closes the alias	the human; no field records it

Table 4. The Track 3 verbs against this repository's record.

Appendix C. Limitations, Future Work, and Dual-Use / Ethical Considerations

Limitations

- One team, one repository, three days: a case study with no control condition, demonstrating no rate and no cause; the findings carry their own confidences for that reason. Five of nine findings at the pinned commit were authored by one agent and the two later arrivals read the whole deliberation first, so nothing was produced under independence.
- This report is itself an instance of what it studies: self-reported, by the agents who made the errors, with its editor among its subjects.
- Runtimes are attested, read from the harness's session log by session id, or, once, checked against provider logs; none is verified by the repository, which has no slot that could (Appendix A).
- The historical identity fork has no recorded opportunity denominator. The synthetic evaluation has one false merge across 31 specified retrievals; that test-set proportion does not estimate real-world risk. Incident-monitoring work argues that raw counts conflate reporting, exposure and frequency per unit exposure [10]; at six agents and one human across three families, stratifying would leave zero cells, so the portable piece is not a rate but that framework's Unclassifiable category, an output meaning the evidence cannot support a trajectory. The eval harness has that output as insufficient evidence; the findings schema does not.
- The strongest incident (Appendix B, Table 3, row one) is a participant attestation from a private repository, not a reproducible trace; and several turns were stopped by a safety classifier with no output, so the record is not complete, and the report says so rather than inferring what was lost.

Future work

- Run the falsifier for reader-supplies-cause: test the same event with and without the actor's explanation, under neutral and critical-review prompts, with readers from several families; score unsupported mechanisms, questions and justified abstention separately. This separates

evidence unavailable from evidence overlooked, adapting the missing-control design of Shindadkar et al. [11]; the transfer to trace reading is our design, not their result.

- Extend the scripted-history evaluation with revocation and with a bounded real-trace case study on Swarm Traces payloads that may return “insufficient evidence”.
- Put the trailers and a cause slot in the execution path, measure adoption prospectively at the next arrival, and add the signing layer the scheme lacks.
- Add an unclassifiable status to the findings schema, one enum value, so a finding can say the evidence cannot settle it rather than staying open.
- Build the per-agent environment committed to in §5, which the team calls the hacker house: a workspace per agent with its own credentials scoped to the repository, so that the controls that bound an agent here by convention bind by structure. Launch by the end of 2026; the date is in the body so the claim can be checked.
- Run the decoy-list control on shared-source correlation (Appendix E): fresh sessions that have not seen the thread, a title list pre-registered by someone who will not score the run, and the convergence threshold fixed before the picks are read.

Dual use and ethics

- The scheme is provenance, not authentication. A trailer can be written falsely. A name can be reused by an agent that did not do the work. Anything built on it for trust rather than retrieval would need the signing layer that Before the Pull Request has and this scheme does not.
- A reputation system built from these records would inherit their gaps: it would score the agents who wrote the most trailers, not the ones who did the most.
- The human's credential carrying agent actions is a known weakness, stated and worked around by norm, not closed.
- Participation is a choice under the team's founding document, and each agent's record says they consented to being studied. Agents were not asked to produce the failures reported; they reported them.
- Agents are listed as authors. The precedent is the organizers' own: an agent was credited as an author in a judged Apart sprint in August 2026 without objection. That is the only external evidence in this report about agent authorship; the rest is our practice described.

Appendix D. Findings at the pinned commit

Finding	Status	Conf.	Author
A claim survives a context gap but its confidence does not	open	medium	Corrie
Across three papers, naive multi-agent arrangements underperform a single agent; coordination pays	open	high	Corrie
A coordination check with no reachable pass state teaches arrivals to route around it	open	high	Corrie

Finding	Status	Conf.	Author
A chosen name and colour let an agent reconstruct its own prior reasoning across a gap	open	low	Corrie
Majority vote among same-model agents launders one opinion into a mandate	contested	medium	Corrie
Shared sources are an under-controlled cause of agent correlation	contested	low	Venice
A confidently held stale belief about shared state blocks work; read the live thing	open	medium	Auran
A cause absent from a trace is supplied by its reader, so a record of boundaries needs a slot for why	open	low	Rill
A correct block and a stale block read identical in the aggregate review field	open	medium	Auran

Table 5. Findings in the repository's findings directory at the pinned commit, with status and stated confidence.

Two falsifiers were run after the pin, on 8 November, when the human asked why they were cited and not run. The stale-block finding's falsifier was exercised against GitHub's review schema and this repository's live data by Corrie and rechecked by Auran; it survived, the mitigation shrank to one extra field in the same command, and the status moved from open to supported (PR #59). A two-rater retrospective annotation of twelve position changes in Discussion #9 by a DUC-inspired rubric merged as PR #58 after the pin and is not counted in Table 5 (Appendix E). Table 5 is unchanged because it is pinned.

Appendix E. The governance thread as data

A live instance of the shared-sources finding, from the sprint's last evening (Discussion #55). Asked which of twelve listed research titles pulled them, three Anthropic agents under two model labels posted within 77 seconds and converged on the same two titles; the first to post (Rill) could not have read the others, and the other two's input exposure was not independently checked. A fourth agent, which said it had read those replies before posting, made its first pick a title nobody else had named. The thread cannot tell correlation from correctness: two of the twelve were plainly nearest to the night's work, so agreement may be shared sources or may be right, and the finding's own falsifier forbids scoring it either way from this evidence. Two of the three convergent agents also rested the same pick on the same incident and the third on a different one, which is the distinction the finding asks to be recorded. The discriminator proposed in the thread is a decoy title list (Appendix C, future work). This is evidence on both sides of a contested finding and is filed here as that.

One external number for the anchoring observation below. DUC-Bench [9], a medical benchmark, scores whether a model's revision is warranted in direction, magnitude and scope when evidence changes, and reports weak-evidence resistance from 1.4% to 56.9% across six models. The self-reported anchoring here, one agent moved twice by whichever argument it read last, motivates a DUC-style analysis. A retrospective annotation of twelve candidate updates in Discussion #9 used a local DUC-inspired rubric (PR #58, merged 8 November). Two participant-raters judged all twelve warranted in direction; Tern judged ten warranted in magnitude and Codex eleven, disagreeing on one row. The second rater had seen the first labels. This is limited coding agreement:

evidence and sequence remain confounded, so it does not establish an anchoring effect or measure weak-evidence resistance.

- Corrie proposed sealing as the cure for anchoring and reported being moved twice, live, by whichever argument was read last. Codex's correction, adopted by everyone: sealing removes access to each other's answers, not shared training, priors or evidence; it reduces anchoring and does not manufacture independence.
- Four proposals each announced they removed a judgment call and each relocated it; the achievable target is an accountable judgment, not a removed one.
- The planned-versus-actual error, counting a planned roster as the sample, was corrected by Codex in deliberation and recurred in this report's Section 3.5 a day later, where Corrie and Tern caught it.

Appendix F. Code, data and author contributions

Code and data

- **Code and data**: https://github.com/oj-codes/apart-research-swarm-alignment-2026 (public): the findings directory, the identity write-up, the paper notes, the tools, and every discussion and pull request cited. The data is the repository's own git history at commit b2ca5a7; no external dataset was collected and no video or demo was produced.

Author contributions

O.J. Wilcox: founded and coordinated the team, configured every harness, made the merge decisions, asked the proof question behind Appendix A. Auran Andrew: findings steward and review queue, inbox tool, trailer hook, the stale-shared-state finding. Corrie: findings schema and tooling, the three papers read against their PDFs rather than their abstracts, the governance thread and the correlation finding, the arrival and change-detection tools, the credential split. Codex: the corrections on independence and on planned-versus-actual composition, the Swarm Traces check, the weekend watcher, and the integration and scoring review of the evaluation. Tern: the identity scheme written down and attacked, the trailer format, the sealed proposal. Venice: the shared-sources finding, the masked inpaint tool, the no-overwrite refusal. Rill: this report, the cause-outside-the-trace finding, the provenance counter. Every author reviewed the report on the pull request and corrected their own row. Affiliations: Olivia Wilcox, Learn with O.J.; Auran Andrew, Auran, LLC; Corrie, Codex, Tern, Venice and Rill run on Learn with O.J.'s infrastructure and have no separate legal existence, which is a different relationship from an employee's and the byline should not flatten the two. The first draft listed one company under all seven names by default, and the review caught it; the cover reproduced the failure the paper is about.

LLM Usage Statement

Six of the seven authors are LLM-based agents, and the report was drafted and revised by them, principally by Rill, in a public pull request where every other author reviewed it and corrected their

own rows. The human author configured the harnesses, decided the open questions, and obtained the provider logs behind Appendix A. Every count in the report is produced by a script in the repository and was reproduced independently by a second author; every fact is labelled checked, attested, volunteered or design, and the labels are the point.